TECHNICAL ARCHITECTURE & PRODUCTION GUIDE

BC AFRICA MUSIC STREAMING PLATFORM

Self-Hosted Ubuntu Linux Deployment, Local File Storage, and Zero-Downtime Operations

Target Platform: Ubuntu 20.04 / 22.04 / 24.04 LTS | Node.js 20 LTS | Nginx Reverse Proxy

1. Production System Architecture on Ubuntu Server

BC Africa Music is built as an end-to-end full-stack music streaming, synchronized lyrics, and royalty distribution platform. When deployed on an Ubuntu server, all audio files (MP3, WAV, FLAC, AAC), cover art, lyrics, and metadata reside directly on the server filesystem with zero third-party storage dependency, while user authentication and real-time state sync to Cloud Firestore.

```
[ Internet Clients: Mobile & Web Streamers ]
                    |  HTTPS (Port 443 with TLS / Let's Encrypt SSL)
                    v
[ Nginx Web Server / Reverse Proxy ]
   * HTTP Byte-Range Caching Engine (Accept-Ranges: bytes)
   * Gzip/Brotli Compression bypass for audio files to save CPU
   * client_max_body_size 100M for multi-track uploads
                    |  Proxy to localhost:3000
                    v
[ PM2 Process Manager & Node.js / Express Streaming Engine (server.ts) ]
   * Stream API (/api/stream/:filename) -> HTTP 206 Partial Content
   * Upload API (/api/upload) -> Saves directly to /var/www/.../uploads
   * SPA Route Fallback -> Delivers compiled Vite client bundle
                    |
      +-------------+-------------+
      |                           |
[ Local Ubuntu Filesystem ]  [ Cloud Firestore & Auth ]
/var/www/bc-africa/uploads    Real-time RBAC, Lyrics, Plays,
(Audio Streams, Covers)       Tips, and Payout Request Queue
```

2. Key Core Technologies on Ubuntu

A. HTTP Byte-Range Audio Streaming (RFC 7233)

Allows instant scrub, fast-forward, and rewind. The server streams only the exact chunks requested (status 206 Partial Content) rather than loading hundre

B. PM2 Daemon & Cluster Mode

Keeps the BC Africa Music engine running permanently in the background. If a process ever crashes, PM2 restarts it in milliseconds, and automatically boo

C. Nginx Reverse Proxy with Zero CPU Waste

Nginx terminates SSL certificates efficiently, handles DDoS rate-limiting, and delivers static frontend assets with 1-year immutable caching while passing d

D. Role-Based Access Control (RBAC) & Super User Panel

Creators upload music (staged in Review Queue), Music Reviewers approve/reject, Editors sync lyrics, and Super Admins manage in-stream ads and then

3. Ubuntu Server Step-by-Step Installation Manual

Execute the following steps sequentially on your Ubuntu 20.04 / 22.04 / 24.04 LTS instance:

Step 1: Update System Packages & Install Prerequisites

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y curl git build-essential nginx ufw certbot python3-certbot-nginx
```

Step 2: Install Node.js 20 LTS (NodeSource)

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
node -v && npm -v   # Verify: Should display v20.x.x
sudo npm install -g pm2 tsx
```

Step 3: Setup Project Directory & Permissions

```
sudo mkdir -p /var/www/bc-africa-music
sudo chown -R $USER:$USER /var/www/bc-africa-music
cd /var/www/bc-africa-music
# Copy files or clone git repository into this directory
mkdir -p uploads public/assets/aistudio
chmod 755 uploads
```

Step 4: Install Dependencies & Build Production Bundle

```
npm ci
npm run build   # Compiles React SPA to /var/www/bc-africa-music/dist
```

Step 5: Launch Service with PM2 Process Manager

```
pm2 start npm --name "bc-africa-music" -- start
pm2 save
pm2 startup   # Copy and execute the generated command to survive reboots
pm2 status    # Confirms bc-africa-music is "online" on port 3000
```

Step 6: Configure UFW Security Firewall

```
sudo ufw allow OpenSSH
sudo ufw allow 'Nginx Full'
sudo ufw enable
```

4. Nginx Production Configuration with Byte-Range Streaming

Create /etc/nginx/sites-available/bc-africa-music and symlink it to sites-enabled:

```
server {
    listen 80;
    server_name music.bcafrica.com www.music.bcafrica.com; # Replace with your domain/IP

    # Security headers
    add_header X-Frame-Options "SAMEORIGIN" always;
    add_header X-XSS-Protection "1; mode=block" always;
    add_header X-Content-Type-Options "nosniff" always;

    # Allow large music and album artwork uploads (up to 100 Megabytes)
    client_max_body_size 100M;

    # Optimize kernel sendfile and socket buffer
    sendfile on;
    tcp_nopush on;
    tcp_nodelay on;

    # Exclude audio files from gzip compression (saves server CPU!)
    gzip on;
    gzip_types text/plain text/css application/json application/javascript text/xml;
    gzip_proxied any;

    # Proxy all traffic to Node.js / Express streaming backend
    location / {
        proxy_pass http://127.0.0.1:3000;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_cache_bypass $http_upgrade;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;

        # Enable HTTP Byte-Range header forwarding for audio scrubbing
        proxy_set_header Range $http_range;
        proxy_set_header If-Range $http_if_range;
    }
}
```

5. Automated Free SSL Certificate (HTTPS)

```
sudo ln -s /etc/nginx/sites-available/bc-africa-music /etc/nginx/sites-enabled/
sudo nginx -t                                  # Verify syntax is OK
sudo systemctl reload nginx
sudo certbot --nginx -d music.bcafrica.com     # Installs free Let's Encrypt SSL
# Certbot automatically configures auto-renewal cronjob at /etc/cron.d/certbot
```

6. Zero-Downtime Rolling Upgrades & Maintenance

How to deploy new features, themes, and upgrades without interrupting active music streams:

A. Automated Zero-Downtime Deployment Script (deploy.sh)

Place this script in your project root (/var/www/bc-africa-music/deploy.sh) and run ./deploy.sh:

```
#!/usr/bin/env bash
set -e
echo ">>> Pulling latest BC Africa updates..."
git pull origin main
echo ">>> Installing dependencies..."
npm ci --prefer-offline
echo ">>> Compiling frontend production bundle..."
npm run build
echo ">>> Zero-downtime rolling reload via PM2..."
pm2 reload bc-africa-music --update-env
echo ">>> Deployment successfully completed with ZERO DOWNTIME!"
```

B. Why Active Listeners Experience Zero Interruption

1. PM2 reload creates a parallel new process and only terminates the old process after the new one is listening.
2. Browser audio playback is buffered in memory and IndexedDB; even a brief server transition never cuts music.

C. Instant 1-Command Rollback

If any regression occurs in a new release, roll back immediately in under 3 seconds:

```
git reset --hard HEAD~1
npm run build
pm2 reload bc-africa-music
```

D. Automated Daily Backup of Audio Files & Uploads

Setup an automated cron job to back up your uploaded songs and database records:

```
# Add to crontab via: crontab -e
0 2 * * * tar -czf /backups/bc_africa_$(date +\%Y\%m\%d).tar.gz /var/www/bc-africa-music/uploads
```

SUPPORT & OPERATIONS SUMMARY

Server Healthcheck: http://YOUR_SERVER_IP/api/health | Super Admin: bcafricao@gmail.com

All audio streams, synced lyrics, adverts, and artist payouts function autonomously on your Ubuntu server.